Users and Permissions

November 13, 2025

In this assignment you will learn how to add users and control read, write and execute permissions. In the first section of the document, we describe the assignment. The subsequent sections give you relevant information to help you complete the task.

1 Assignment

In figures 1, 2 and 3 you will find an image of the directory tree and the checksums of the files in the tree. Note that figure 3 is the same as 2, just zoomed in so it's easier to read. Your job is to recreate this tree with the proper permissions and checksums.

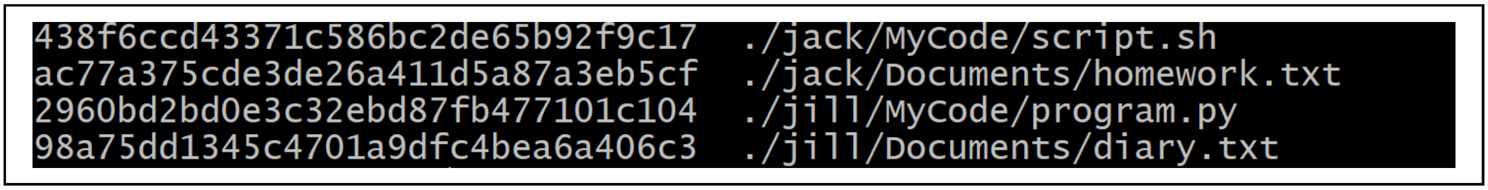
```
root@droplet:/home# tree -p
[drwxr-xr-x]  .
├── [drwx------]  jack
│   ├── [drwxr-xr-x]  Documents
│   │   └── [-rw-r--r--]  homework.txt
│   └── [drwx------]  MyCode
│       └── [-rwx------]  script.sh
└── [drwx------]  jill
    ├── [drwxrwxr-x]  Documents
    │   └── [-rw-------]  diary.txt
    └── [drwxrwxr-x]  MyCode
        └── [-rwxr-xr-x]  program.py

7 directories, 4 files
```

Figure 1: `Reproduce this tree`

```
root@droplet:/home# find . -type f \( -name *.txt -o -name *.py -o -name *.sh \) | xargs -I {} -n1  md5sum {} ;
438f6ccd43371c586bc2de65b92f9c17  ./jack/MyCode/script.sh
ac77a375cde3de26a411d5a87a3eb5cf  ./jack/Documents/homework.txt
2960bd2bd0e3c32ebd87fb477101c104  ./jill/MyCode/program.py
98a75dd1345c4701a9dfc4bea6a406c3  ./jill/Documents/diary.txt
```

Figure 2: `Make sure the checksums match.`

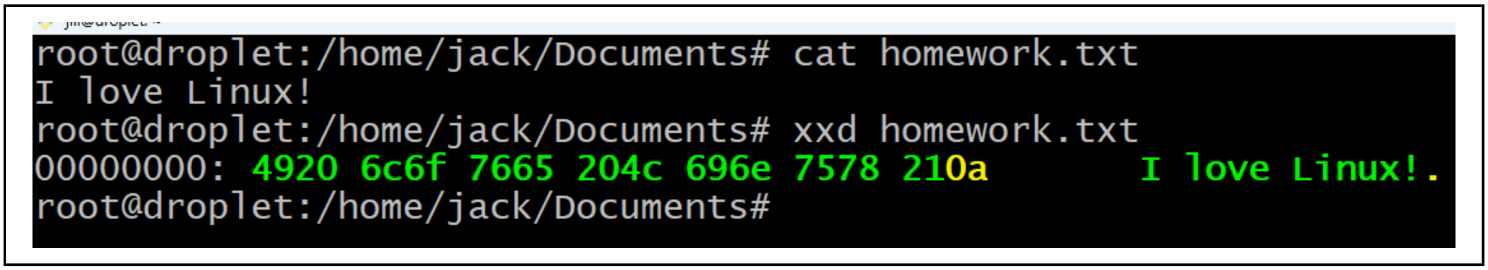


```
438f6ccd43371c586bc2de65b92f9c17  ./jack/MyCode/script.sh
ac77a375cde3de26a411d5a87a3eb5cf  ./jack/Documents/homework.txt
2960bd2bd0e3c32ebd87fb477101c104  ./jill/MyCode/program.py
98a75dd1345c4701a9dfc4bea6a406c3  ./jill/Documents/diary.txt
```

Figure 3: `Closeup of checksums so it's easier to read.`

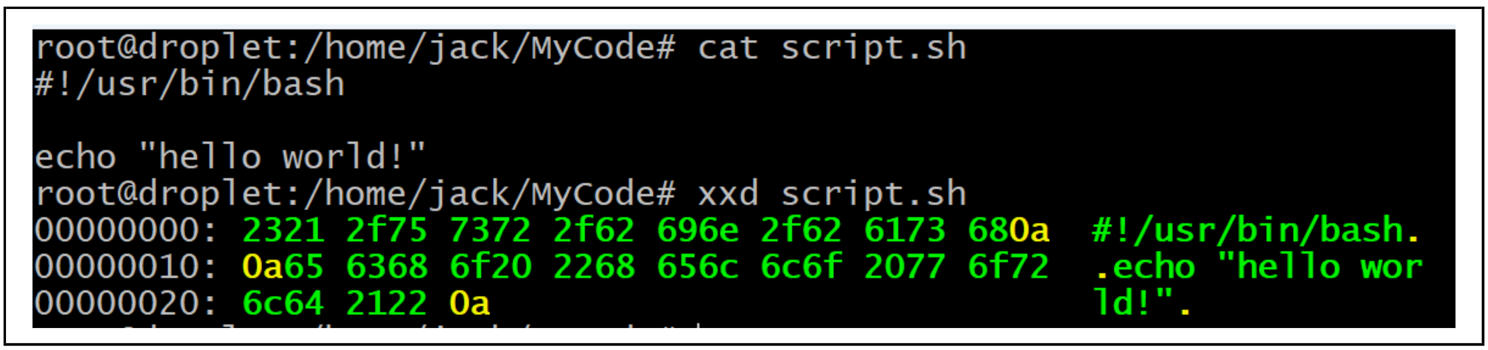
1.1 The Files

In figures 4, 5, 6 annd 7 you will find information about the contents of the files.



```
root@droplet:/home/jack/Documents# cat homework.txt
I love Linux!
root@droplet:/home/jack/Documents# xxd homework.txt
00000000: 4920 6c6f 7665 204c 696e 7578 210a      I love Linux!.
root@droplet:/home/jack/Documents#
```

Figure 4: `/home/jack/Documents/homework.txt`



```
root@droplet:/home/jack/MyCode# cat script.sh
#!/usr/bin/bash

echo "hello world!"
root@droplet:/home/jack/MyCode# xxd script.sh
00000000: 2321 2f75 7372 2f62 696e 2f62 6173 680a  #!/usr/bin/bash.
00000010: 0a65 6368 6f20 2268 656c 6c6f 2077 6f72  .echo "hello wor
00000020: 6c64 2122 0a                             ld!".
```

Figure 5: `/home/jack/MyCode/script.sh`

```
root@droplet:/home/jill/Documents# ls
diary.txt
root@droplet:/home/jill/Documents# cat diary.txt
today was a good day.
root@droplet:/home/jill/Documents# xxd diary.txt
00000000: 746f 6461 7920 7761 7320 6120 676f 6f64  today was a good
00000010: 2064 6179 2e0a                            day..
```

Figure 6: `/home/jill/Documents/diary.txt`

```
root@droplet:/home/jill/MyCode# cat program.py
#!/usr/bin/python3

import this
root@droplet:/home/jill/MyCode# xxd program.py
00000000: 2321 2f75 7372 2f62 696e 2f70 7974 686f  #!/usr/bin/pytho
00000010: 6e33 0a0a 696d 706f 7274 2074 6869 730a  n3..import this.
root@droplet:/home/jill/MyCode# |
```

Figure 7: `/home/jill/MyCode/program.py`

2 Install programs

In this assignment you'll be using the **tree** and **xxd** programs. Install them like:

```
root@droplet$ apt install tree xxd
```

- **tree** - lets you see your directory structure as a nice graphical tree.
- **xxd** - gives a hex dump of a file so you can see the file in hexadecimal. This is an important command you'll use when inspecting files. We present it in this docuument, which is likely your first exposure to the xxd command.

3 Creating the tree

3.1 users

Linux is a multi-user operating system. Our droplets by default only come with one user - root - but you can add more users to your computer.

3.2 adduser

To add a user, you use the **adduser** command. When you add a user you'll be prommpted to give a password. After providing the password, you're asked a bunch of questions. You can just hit ENTER repeatedly and not enter values for things like Room Number.

See figure 8. Note that in the beginning there are only directories for jack and jill. When I add the user bruno and **ls /home**, you see that there is now a home directory for bruno. When you run **adduser**, a home directory is added for the user.

```
root@droplet:~# ls /home
jack  jill
root@droplet:~# adduser bruno
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bruno
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
root@droplet:~# ls /home
bruno  jack  jill
root@droplet:~#
```

Figure 8: `adduser`

3.3 groups

Linux users can be binned into groups. This is an important topic, but will be left for a future lecture. If we start discussing groups in this lecture, it might feel like too much material.

3.4 addgroup

You can add groups to your computer with **addgroup**. But, as mentioned above, that is outside of the scope of today's assignment.

3.5 File ownership

Every file in Linux has a user owner and a group owner. See figure 9. The output of **ls -l** shows who is the user owner and who is the group owner of any file or directory.

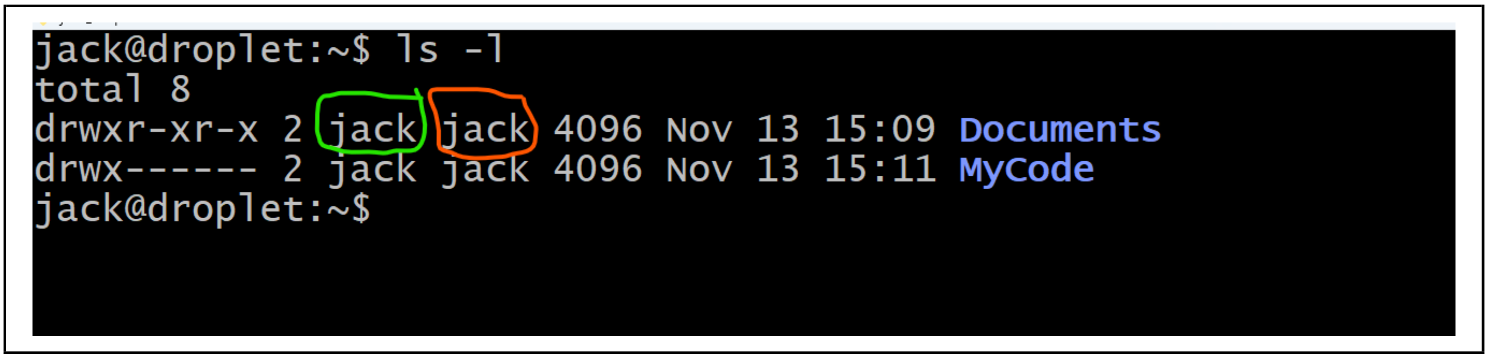


Figure 9: `See the output of ls -l. The user owner if highlighted in green, and group owner is highlighted in orange.`

3.6 chown

To change the owner of a file you can use chown. If figure 10 you see how to change the file owner. If figure 11 you see how to change the owner and the group. In figure 12 you see how to change ownership recursively for all of the contents of a directory.

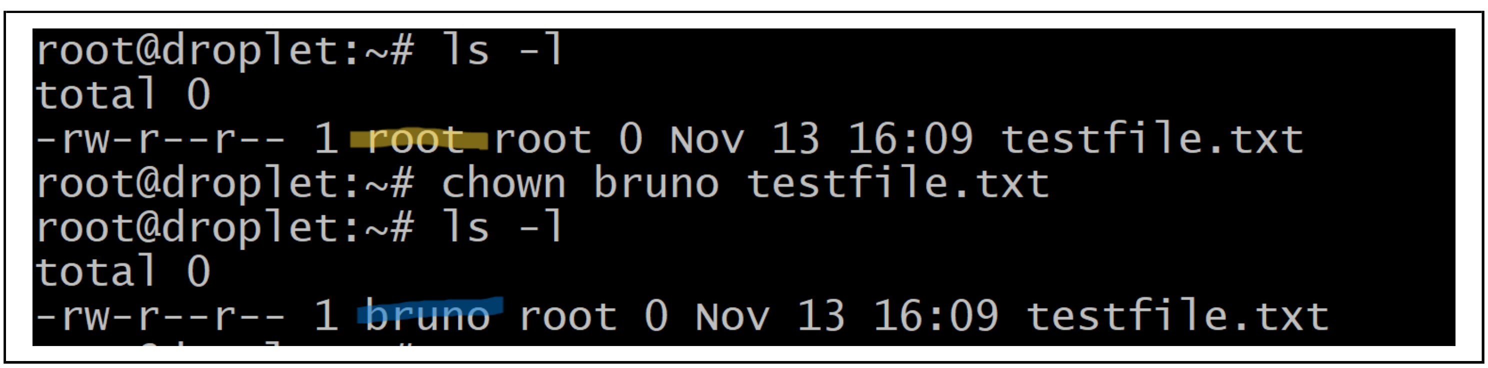
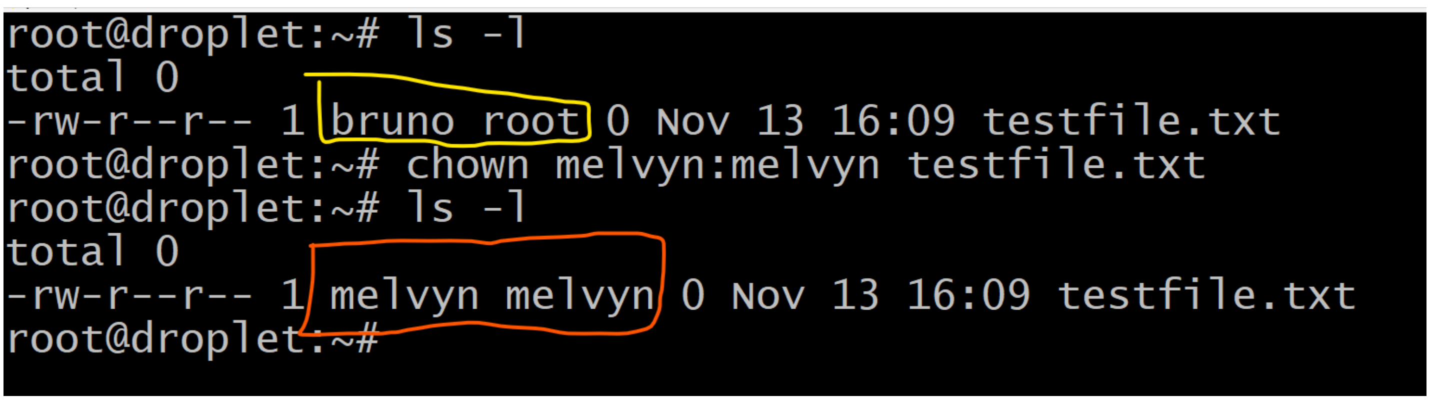
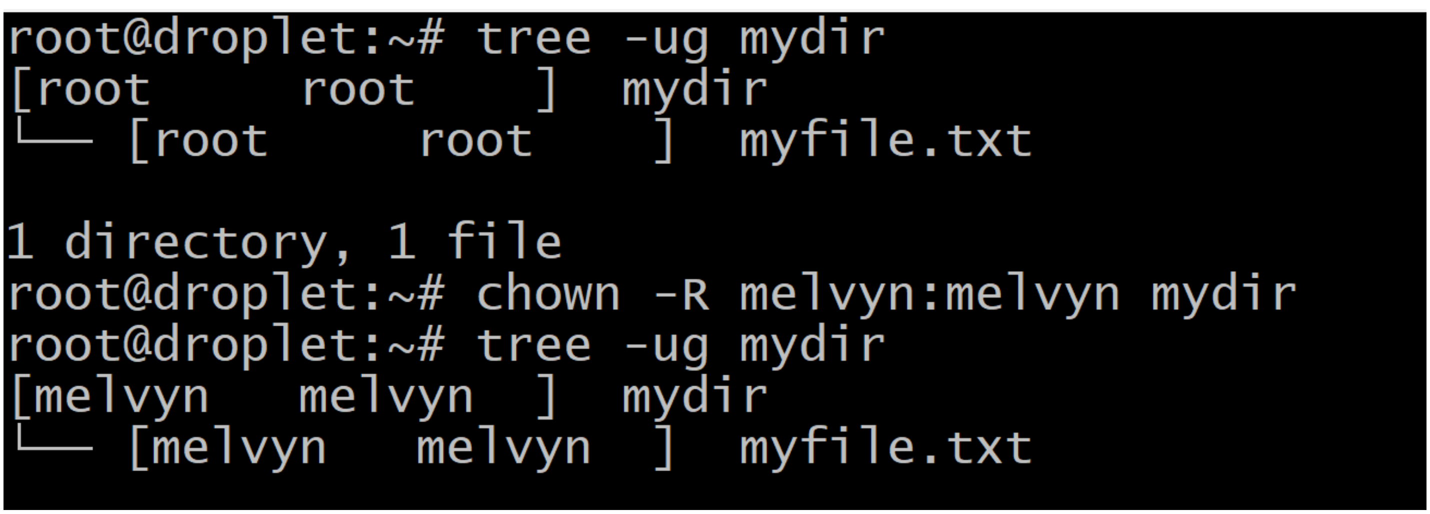


Figure 10: `See highlighted in yellow that the owner was root. After running the chown command, the owner is bruno.`



```
root@droplet:~# ls -l
total 0
-rw-r--r-- 1 bruno root 0 Nov 13 16:09 testfile.txt
root@droplet:~# chown melvyn:melvyn testfile.txt
root@droplet:~# ls -l
total 0
-rw-r--r-- 1 melvyn melvyn 0 Nov 13 16:09 testfile.txt
root@droplet:~#
```

Figure 11: See that the user/group of the file was bruno/root. But after chown, the user/group are melvyn/melvyn.



```
root@droplet:~# tree -ug mydir
[root     root    ]  mydir
└── [root     root    ]  myfile.txt

1 directory, 1 file
root@droplet:~# chown -R melvyn:melvyn mydir
root@droplet:~# tree -ug mydir
[melvyn   melvyn  ]  mydir
└── [melvyn   melvyn  ]  myfile.txt
```

Figure 12: Instead of using ls -l, I use tree -ug so it's easier to see the ownership of a directory and its contents. We use chown with the -R flag to change the ownership of a directory and its contents.

3.7 Permissions

Linux files and directories have permissions. Files can directories can be:

- readable
- writable
- executable

These permissions are specified in three ways:

- owner
- group
- everyone else

So files and directories have 9 permission fields. They have readable, writable and executable for each of owner, group and everyone else. See figures ?? and ?? to see two examples of permissions. Permissions are show with 9 characters.

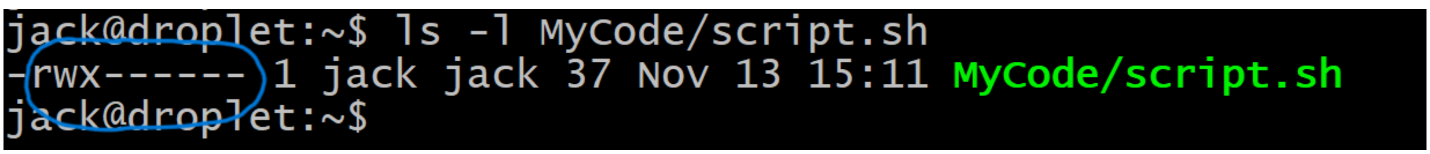


Figure 13: `See the circled area. 9 characters indicate read, write an execute for each of owner, group, and everyone else`

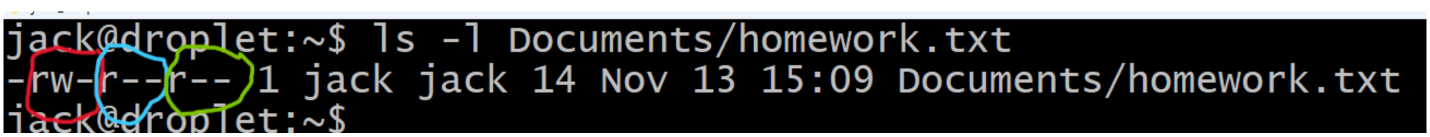


Figure 14: `The red circle indicates permissions for owner. Blue indicates group. Green indicates everyone else`

3.8 chmod

4 The Checksums